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Adopting concept of biophilic design to enhance healthy indoor: an experiment on moss-base for tropical building



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Abstract

The modern world is increasingly focused on sustainability, promoting the concept of a greener Earth. This forms the foundation for Industry 6.0, which aims to harness emerging technologies in harmony with nature, supporting sustainability principles where technology complements rather than dominates the environment. Within this framework, there is growing interest in exploring Nature-Based Solutions (NbS) for buildings, aiming to enhance indoor performance. The concept of integrating nature with the built environment is referred to as Biophilic Design (BD). It encompasses a variety of strategies, among which the incorporation of plants is prominent as a widely adopted practice in industry. Numerous indoor plants, such as Spider Plant, Peace Lily, Areca Palm, Rubber Plant, Aloe Vera, Pothos (commonly known as Money Plant), and non-vascular plants, such as bryophyte species, are widely used in BD. This study specifically investigates the efficacy of integrating moss plants (a type of bryophyte species), which are abundant in tropical countries like Sri Lanka, due to their ability to improve indoor air quality (IAQ). An experimental research method was employed to assess the effectiveness of moss plants in enhancing the IAQ. The experiment conducted under a controlled environment proved that Moss plants effectively reduced particulate matter (PM10) and removed formaldehyde. These findings establish a foundation for future research to develop this concept into a commercially practical product, using moss as a readily available resource in Sri Lanka.

Keywords

[Moss \(Bryophyte\)](#), [IAQ](#), [Biophilic Design](#), [Indoor Air Quality](#), [Healthy Indoor](#), [Tropical Environment](#)

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